

Lab 01: Build a responsive personal portfolio page using semantic HTML5, CSS grid/flexbox, and media queries

Objectives:

- a) To design a simple personal portfolio webpage using HTML.
- b) To apply CSS for styling, layout, and visual presentation of a web page.
- c) To use flexbox for creating responsive layouts.
- d) To implement responsive design using media queries for different screen sizes.

Theory

HTML (HyperText Markup Language)

It is the standard markup language used to create the basic structure of web pages. It organizes content using elements such as headings, paragraphs, images, links, and semantic tags like header, navigation, main section, and footer. HTML tells the browser what content to display and how different parts of a webpage are organized. By using clear and meaningful tags, HTML helps create webpages that are easy to read, navigate, and understand for both users and search engines.

CSS (Cascading Style Sheets)

It is used to control the appearance and visual design of web pages. It allows developers to add colors, fonts, spacing, layouts, and animations to HTML content, making websites more attractive and user-friendly. CSS can be applied in three different ways: inline CSS, which is written directly inside HTML elements; internal CSS, which is placed within a <style> tag in the HTML document's head section; and external CSS, which is written in a separate stylesheet file and linked to the HTML page.

Responsive Web Design

Responsive web design ensures that a webpage looks good on all devices, including desktops, tablets, and smartphones. This is achieved using flexible layouts, relative units, and media queries.

1. Flexbox (Flexible Box Layout)

It is a CSS layout model designed to make it easier to arrange and align elements within a container. It allows items to automatically adjust their size and position based on available space, making layouts more flexible and responsive

2. Media queries

They are a feature of CSS that allow web pages to adapt their layout and design based on different screen sizes and devices. They enable developers to apply specific styles when certain conditions, such as screen width, height, or orientation, are met.

Discussion

This lab enhanced the understanding of how HTML and CSS are used together to build well-structured web pages. It demonstrated how CSS styling improves the visual appearance and usability of a website. The use of Flexbox made layout alignment more flexible and efficient. Media queries helped the webpage adapt to different screen sizes and devices.

Conclusion

In conclusion, this lab provided a strong foundation in understanding the basics of web development using HTML and CSS. It helped demonstrate how proper structure, styling, and layout techniques contribute to creating visually appealing and responsive web pages.